Class: Grade 12 LS **Subject: Mathematics**

Teacher: Muhieddeen Iskandarani **Extra Sheet: Continuous function over an interval**

Exercise 1.

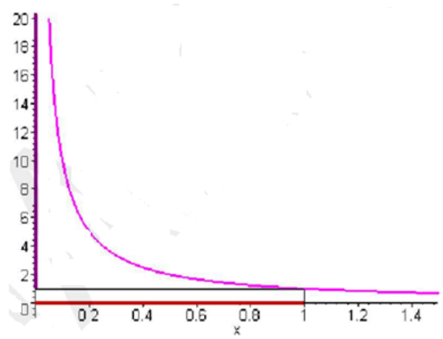
Consider the table of variations of a function f.

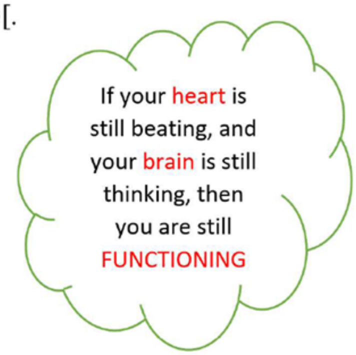
x	$-\infty$		$-\frac{5}{2}$		$-\frac{1}{2}$		2		$+\infty$
$f'(x)$		$-$		$-$	0	$+$		$+$	
$f(x)$	$+\infty$	$\searrow$	$\frac{7}{4}$	$\searrow$	$-\frac{9}{4}$	$\nearrow$	4	$\nearrow$	$+\infty$

Determine the image by f of each of the following intervals: $\left]-\infty;-\frac{5}{2}\right]$, $\left[-\frac{1}{2};2\right]$ and $\left[-\frac{5}{2};2\right]$.

Exercise 2.

Consider the function $f(x)=\frac{1}{x}$ and its representative curve over $]0;+\infty[$.





Determine: a) f(]0: 1]) ; b) f(]0; 1[) ; c) f(]0; 0.5]) ; d) f(]0; 2[)

Exercise 3:

Consider the table of variations of a function f.

x	$-\infty$		0		2		4		$+\infty$
Variations of f	$+\infty$	$\searrow$	5	$\nearrow$	$+\infty$ || $-\infty$	$\nearrow$	3	$\searrow$	-5

1) Determine $f(]2;+\infty[)$.
2) Show that the equation $f(x)=0$ has no real roots.

Exercise4:

Consider the table of variations of a function f.

x	$-\infty$		-3		2		$+\infty$
Variations of f			126				$+\infty$
		$\nearrow$		$\searrow$		$\nearrow$	
	$-\infty$				1		

1) Show that the equation $f(x) = 0$ has a unique real root.
2) Show that the equation $f(x) = 2$ has exactly 3 real roots.

Exercise 5:

Consider the table of variations of a function f.

x	-3		1		4		7
$f(x)$	7				-1		
		$\searrow$		$\nearrow$		$\searrow$	
			-3				-2

1) Determine $f([-3; 7])$.
2) Show that the equation $f(x) = 0$ has a unique solution.

Exercise 6:

1) Show that the equation $x^8 + 8x^3 + x - 3 = 0$ admits at least one real root between 0 and 1.
2) Determine the number of real solutions of the equation $x^3 - 3x^2 - 1 = 0$.
3) Consider the function $f(x) = \frac{x^3}{x+1}$.
   a) Draw the table of variations of f over $[-1.5; -1[$.
   b) Show that the equation $f(x) = 10$ admits a unique root in the interval $[-1.5; -1[$.
4) Consider the function $f(x) = \sqrt{x^2 + 1}$.
   a) Draw the table of variations of f over $]-\infty; 0]$.
   b) Show that the equation $f(x) = 2$ admits a unique root in the interval $]-\infty; 0]$.

Exercise 7:

A- Let g be the function defined on IR by $g(x) = 2x^3 + ax^2 - 3$, where a is real constant, let (Γ) be its representative curve in an orthonormsal system$(O, \vec{i}, \vec{j})$.

1) Knowing that the tangent to the curve (Γ) at the point of abscissa 1 is parallel to the line of equation y = 12x, show that a = 3.
2) Study the variations of g and set up its table of variations.
3) Show that the equation g (x) = 0 admits on IR a unique root α. Verify that $0.8 < \alpha < 0.9$.
4) Show that the curve (Γ) has a point of inflection I whose coordinates are to be determined.
5) Draw the curve (Γ).
6) Study the sign of g (x) according to the values of x.

B- Let f be the function defined over]-1 ; +∞[by $f(x) = \frac{x^3+3}{x+1}$, and let (C) be its representative curve in an orthonormal system $(O, \vec{i}, \vec{j})$.

1) Calculate $\lim_{x \to -1} f(x)$. Interpret the result graphically.
2) Calculate $\lim_{x \to +\infty} f(x)$.
3) Prove that $f'(x) = \frac{g(x)}{(x+1)^2}$, for all x>-1, then set up the table of variations of f.
4) Let α=0.85, Trace (C).
5) Discuss graphically, according to the values of the real parameter m, the number of roots of the equation f (x) = m.